TaskHive

A Software for Maximizing Efficiency with Hive-Like Precision

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Project Plan

Sponsor: N/A

Project Name: TaskHive

Project Scope

TaskHive is a task management system designed to optimize productivity and streamline workflows. It enables users to create, assign, and monitor tasks efficiently while maintaining control over priorities, deadlines, and progress tracking from start to finish. With support for recurring tasks, the system ensures that repetitive tasks are handled automatically, while notifications keep users informed of approaching deadlines. It also offers detailed progress reports, providing insights into the task completion and overall productivity, making it a comprehensive solution for efficient organization and task management. Additionally, the system includes user profile management and role-based permissions to ensure secure and organized task handling.

Key Features

1. **Task Creation and Management:**
    - Allows users to create, edit, and delete tasks.
    - Tasks can be categorized under “Work”, “Personal”, and custom labels.

- Tasks will have fields for priority, due dates, descriptions, and tags.
- Allow users to set tasks that repeat either daily, weekly, monthly, or yearly.

2. **Insights and Reports**
   - Visual representation of progress tracking with percentage completions and graphs.
3. **Notifications and Alerts**
   - Set automatic reminders upcoming tasks.
   - Push notifications for deadlines, team updates, priority tasks.
4. **User Authentication and Profile Management**
   - Track individual productivity over time.
   - Set personal goals for using application.
5. **Integration**
   - Sync with calendar applications (Google, Apple, Outlook).
   - Integrate email platforms, cloud storage, and project management apps.

System Architecture Overview

Software Requirements

For the software requirements of this project, the primary operating system that will be utilized is Windows 11 (64-bit). This environment provides compatibility with modern development tools and supports the use of Docker with the WSL2 backend. The main development tools used for this project includes Visual Studio Code, Docker Desktop, PostgreSQL, Redis FastAPI, Celery, and Node.js with Vite React TypeScript.

Visual Studio Code will serve as the primary development environment and text editor for both backend and frontend development. It will be configured with extensions such as Python and Pylance to assist with linting, formatting, and code navigation. Docker Desktop will be used to manage containerized services, including the PostgreSQL database, Redis message broker, and the FastAPI backend. PostgreSQL will be used as the main relational database for storing application data, while Redis will serve as a message broker to support background task processing with Celery.

On the backend, the project will utilize Python 3.11+ with FastAPI as the primary web framework, Uvicorn as the ASGI server, and Alembic for handling database migrations. The frontend will be developed using Node.js (version 18 or 20) and Vite to build a modern React + TypeScript single-page application. Git will be used for version control, and PowerShell within Windows Terminal will be used to execute setup and deployment commands.

The expected runtime environment will expose services through standard development ports — the FastAPI backend will run on port 8000, the React frontend on port 5173, PostgreSQL on 5432, and Redis on 6379.

Hardware Requirements

For the hardware requirements, this project will require a development machine capable of running Docker, Visual Studio Code, and other development dependencies efficiently. The system should include at least a quad-core processor such as an Intel i5 or AMD Ryzen 5, with a minimum of 8 gigabytes of RAM. However, 16 gigabytes of RAM or more is recommended to ensure smooth operation when running multiple Docker containers and local development servers concurrently.

A minimum of 20 gigabytes of available disk space is required for storing Docker images, database volumes, and project dependencies, though an SSD with at least 30–50 gigabytes of free space is recommended for optimal performance. The system should also support hardware virtualization (Intel VT-x or AMD-V) to enable WSL2 for Docker. Finally, a stable internet connection is needed for installing dependencies, pulling Docker images, and accessing external development libraries.

Requirements

Functional Requirements

FR1 — Task Creation and Management

- FR1.01: The system must allow users to create a new task and set its name, description, custom labels, deadline, priority, and recurrence .
- FR1.02: The system must allow users to set task's recurrence to repeat daily, weekly, monthly, or yearly.
- FR1.03: The system must allow users to categorize tasks using custom labels or tags for better organization
- FR1.04: The system must allow users to edit all details of an existing task.
- FR1.05: The system must allow users to delete tasks and undo deletes.
- FR1.06: The system shall track the status of tasks as "to-do", "in-progress", or "completed".
- FR1.07: All users must assign a name, deadline, and priority level to tasks to complete the task creation process.
- FR1.08: The system shall provide a visual calendar interface that displays tasks with associated deadlines and milestones. The calendar shall allow users to view tasks by day, week, and month, ensuring a clear overview of upcoming tasks and deadlines.
- FR1.09: The system shall allow users to plan and block specific time slots for tasks. Users must be able to select time ranges and assign them to particular tasks, ensuring effective time management and task scheduling.

FR2 —Insights and Reports

- FR2.01: The system shall generate reports and present them as dashboard overviews, featuring progress bars and summaries that reflect task completion, productivity, and timelines.
- FR2.02: The system shall allow users to track the time spent on individual tasks

FR3 — Notifications and Alerts

- FR3.01: The system must allow users to set notifications for assigned tasks' upcoming deadlines.
- FR3.02: The system shall allow users to customize notification preferences (e.g., email, SMS, or in-app).
- FR3.03: The system shall allow users to set multiple reminders for a task (e.g., one day before, one hour before, etc.).
- FR3.04: The system shall notify users for tasks marked as high priority.
- FR3.05: The system must allow users to disable notifications when currently enabled.

FR4 — User Authentication and Profile Management

- FR4.01: All users must create a profile by providing an email and password.
- FR4.02: All users must validate their profile via email
- FR4.03: All users must login using a valid username or email and a password to access the system
- FR4.04: The system shall verify user credentials upon login.

- FR4.05: The system must allow users to delete their account by password verification and email confirmation.
- FR4.06: The system must provide a password recovery option for users.
- FR4.07: The system must lock user's account after five failed login attempts.

FR5 — Integration

- FR5.01: The system shall enable the option for synchronizing tasks with users' external calendar applications in real-time.
- FR5.02: The system shall integrate with external tools lie email platforms, cloud storage, and project management apps (e.g., Microsoft Teams, Google Drive, etc.).

Nonfunctional Requirements

NFR1—Efficiency Requirements

- NFR1.01: Task data, including images, must be compressed to reduce storage needs.
- NFR1.02: The system must not consume more than 5% of battery life during each usage.

NFR2—Dependability Requirements

- NFR2.01: The system must back up all user data daily, with the ability to restore data within 10 minutes in the event of operational disruption.
- NFR2.02: Maintenance timeframes shall not exceed more than 1 hour per month, and user must be notified 24 hours in advance.
- NFR2.03: The system must display user-friendly error messages that are concise and directly relate to the issue that occurred, and solutions or recovery options in less than 5 seconds after an error occurs.
- NFR2.04: The system shall notify the developers of any critical errors within 1 minute of detection

NFR3 — Security Requirements

- NFR3.01: The system shall lock user accounts after 5 failed login attempts within a 10-minute timeframe
- NFR3.02: All user data must be encrypted
- NFR3.03: The system must provide multi-factor authorization as an advanced security option for all users to secure their accounts.
- NFR3.04: The system must automatically log users out after 15 minutes of inactivity to prevent unauthorized access.

NFR4 — Performance requirements

- NFR4.01: The system must respond within 3 seconds when a user creates, edits, or deletes a task

NFR5 — Usability Requirements

- NFR5.01: The system shall allow new users the option to complete a tutorial to acclimate to the system's usage or bypass the training
- NFR5.02: The user interface must be accessible and optimized for both mobile and desktop platforms

NFR6 — Environmental requirements

- NFR6.01: The system should be compatible with all major operating systems (Windows, macOS, Linux).

NFR7 — Space Requirements

- NFR7.01: The system must not require more than 1 GB of storage per user

NFR8 — System requirements

- NFR8.01: The system shall provide backups every 24 hours to ensure data recover in case of failures

NFR9 — Ethical requirements

- NFR9.01: The system must clearly state how user data is collected and used to ensure transparency

Project Modularization

1. Task creation and management

- Create Task: Create new task (initially assigned "to-do' as a default state)
  - Input: Task details (name, description, deadline, priority level (1-3), recursion setting).
  - Output: Task created, added to task list, and assigned "to-do" status.
- Edit tasks: Modify and update an existing task's details.
  - Input: Task details (name, description, deadline, priority level (1-3), recursion setting).
  - Output: Modified task details
- Delete tasks: Erases unwanted tasks from users' list of tasks
  - Output: Task deleted from list
- Task Progress Update ("In-Progress"): Users can manually change a task's status to "in-progress" when they start working on it.
  - Input: User action to mark as started.
  - Output: Task status changes from "To-Do" to "In-Progress".
- Task Completion ("Completed")
  - Input: User action to mark task as completed, along with possible checks for dependencies.
  - Output: Task stats changes from "In-Progress" to "Completed".
- Automated Checks for Task Status: Automated checks to track task status

- Tasks Status Reporting: Monitor progress of tasks and generate a dashboard overview based on reports
- Set Prioritization: Set priority level (scale of 1-3); Notify for high priority tasks
- Task Recursion: Set task that repeat on daily, weekly, or monthly

2. **User Management**
   - New User: Create profile
     - Input: Set email address, password, and profile image
     - Output: Profile created
   - Manage Profile
     - Delete profile
     - Change Email Address
     - Change Password
   - User Validation: Verify user credentials

3. **Notifications**
   - Send Notifications

4. **Integration**
   - Allow user to sync the system with calendar applications

User Interface — Wireframes

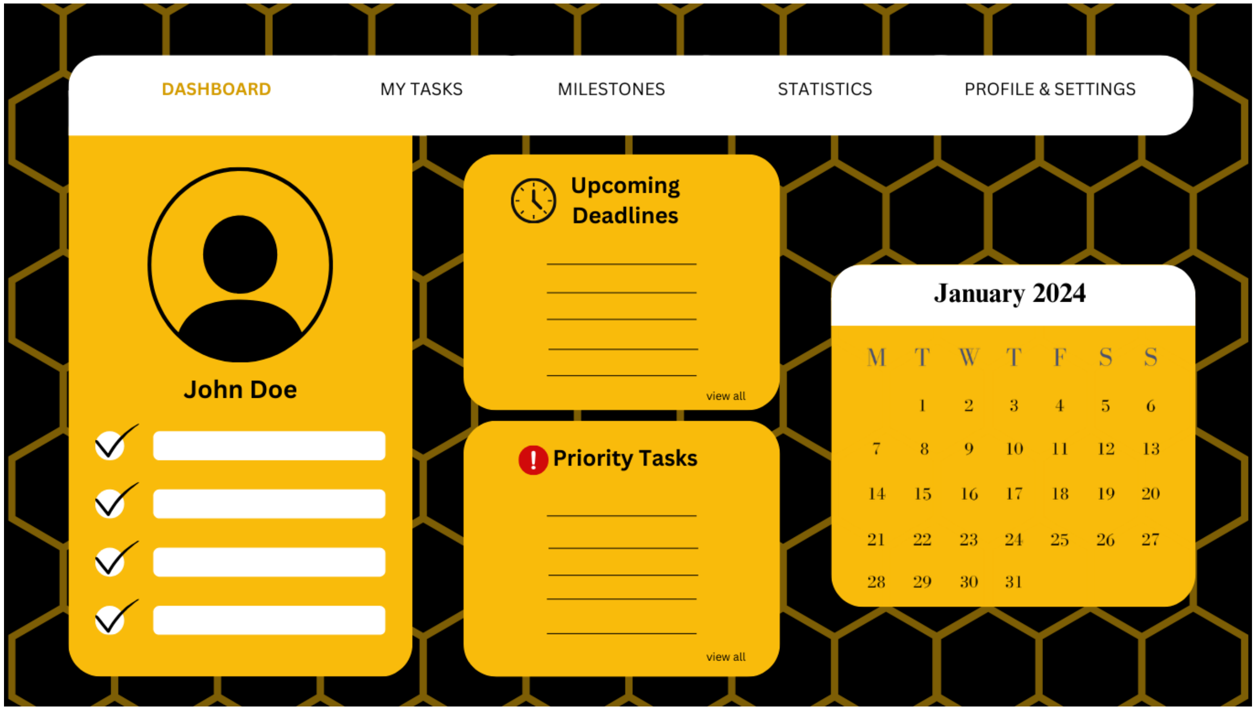


Figure 1

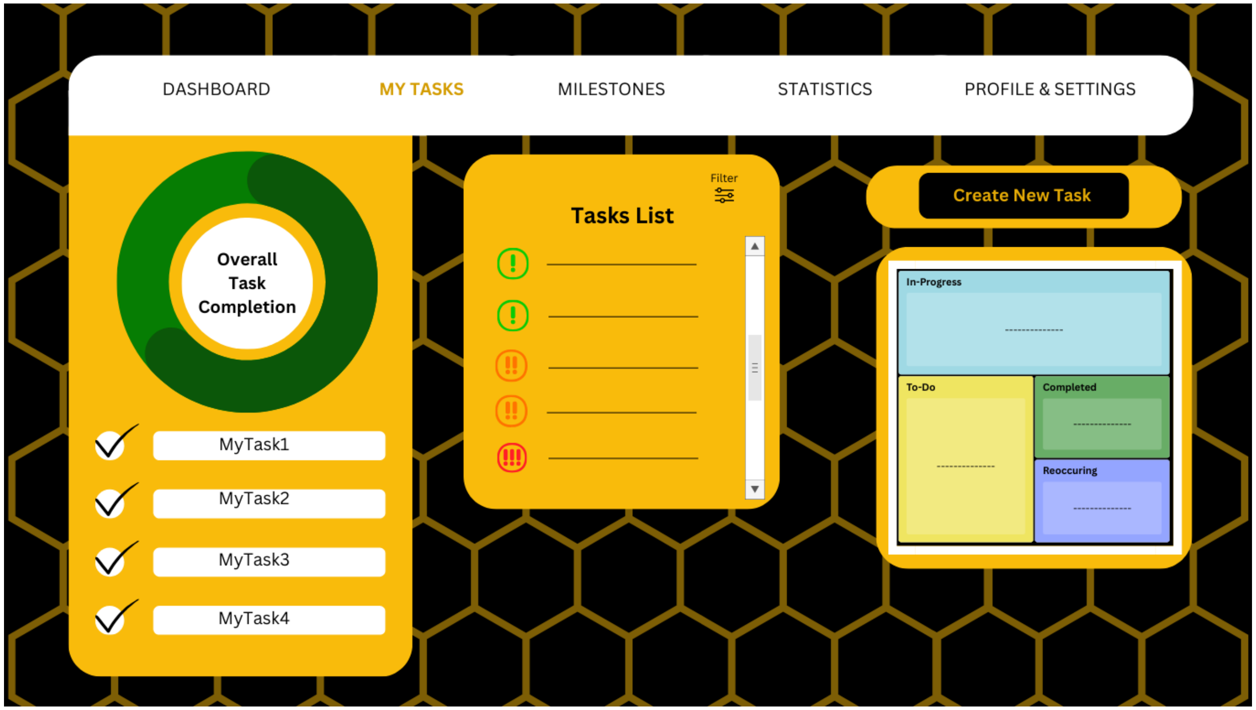

DASHBOARD
MY TASKS
MILESTONES
STATISTICS
PROFILE & SETTINGS
Overall
Task
Completion
MyTask1
MyTask2
MyTask3
MyTask4
Filter
Tasks List
Create New Task
In-Progress
To-Do
Completed
Reoccuring

Figure 2

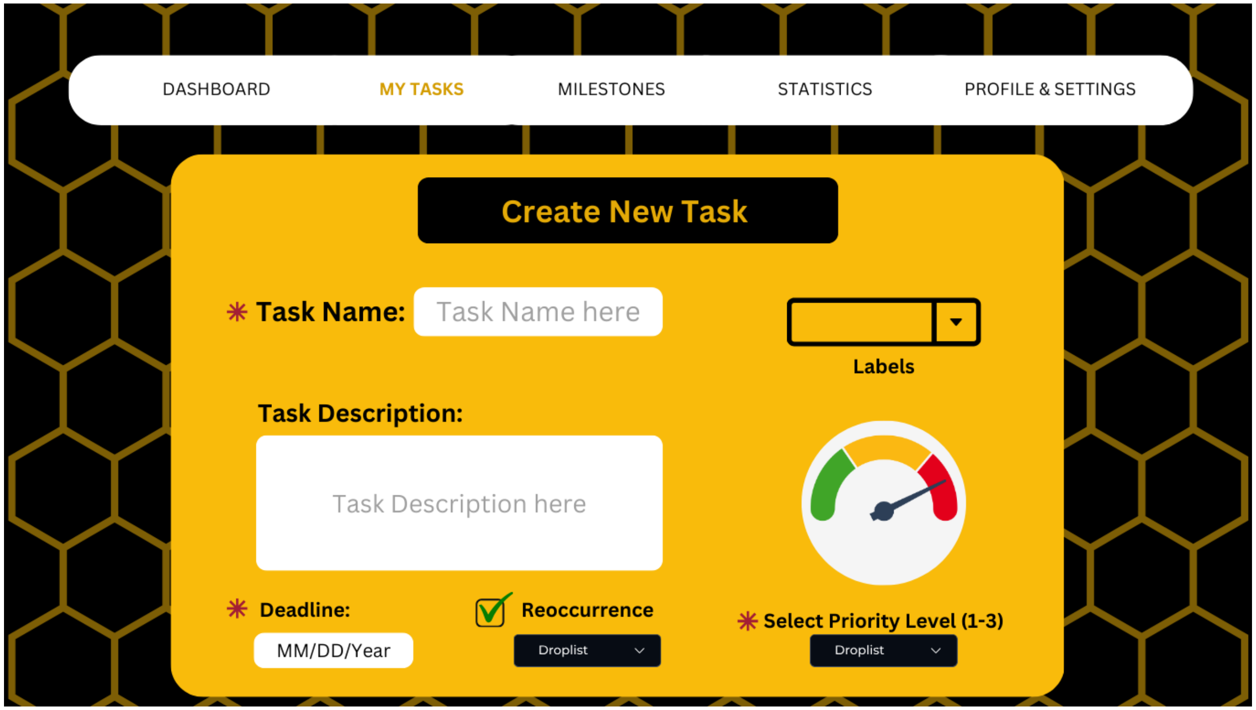

DASHBOARD
MY TASKS
MILESTONES
STATISTICS
PROFILE & SETTINGS
Create New Task
Task Name:
Task Name here
Labels
Task Description:
Task Description here
Deadline:
MM/DD/Year
Reoccurrence
Droplist
Select Priority Level (1-3)
Droplist

Figure 3

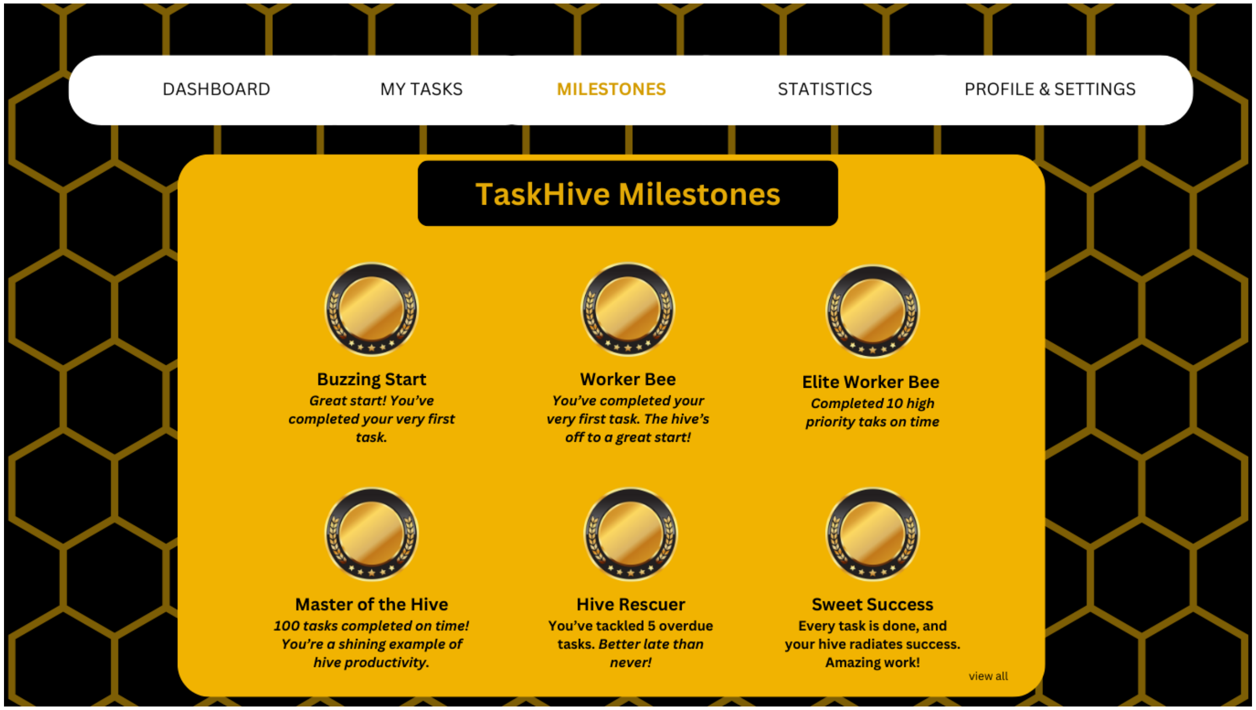

DASHBOARD
MY TASKS
MILESTONES
STATISTICS
PROFILE & SETTINGS
TaskHive Milestones
Buzzing Start
Great start! You've completed your very first task.
Worker Bee
You've completed your very first task. The hive's off to a great start!
Elite Worker Bee
Completed 10 high priority taks on time
Master of the Hive
100 tasks completed on time! You're a shining example of hive productivity.
Hive Rescuer
You've tackled 5 overdue tasks. Better late than never!
Sweet Success
Every task is done, and your hive radiates success. Amazing work!
view all

Figure 4


DASHBOARD
MY TASKS
MILESTONES
STATISTICS
PROFILE & SETTINGS
Statistics & Reports
Click to view...
Task Completion Metrics
Task Prioritization & Deadlines
Task Type & Category Breakdown
Task Dependency & Bottlenecks
Delay Analysis
Milestone Tracking

Figure 5

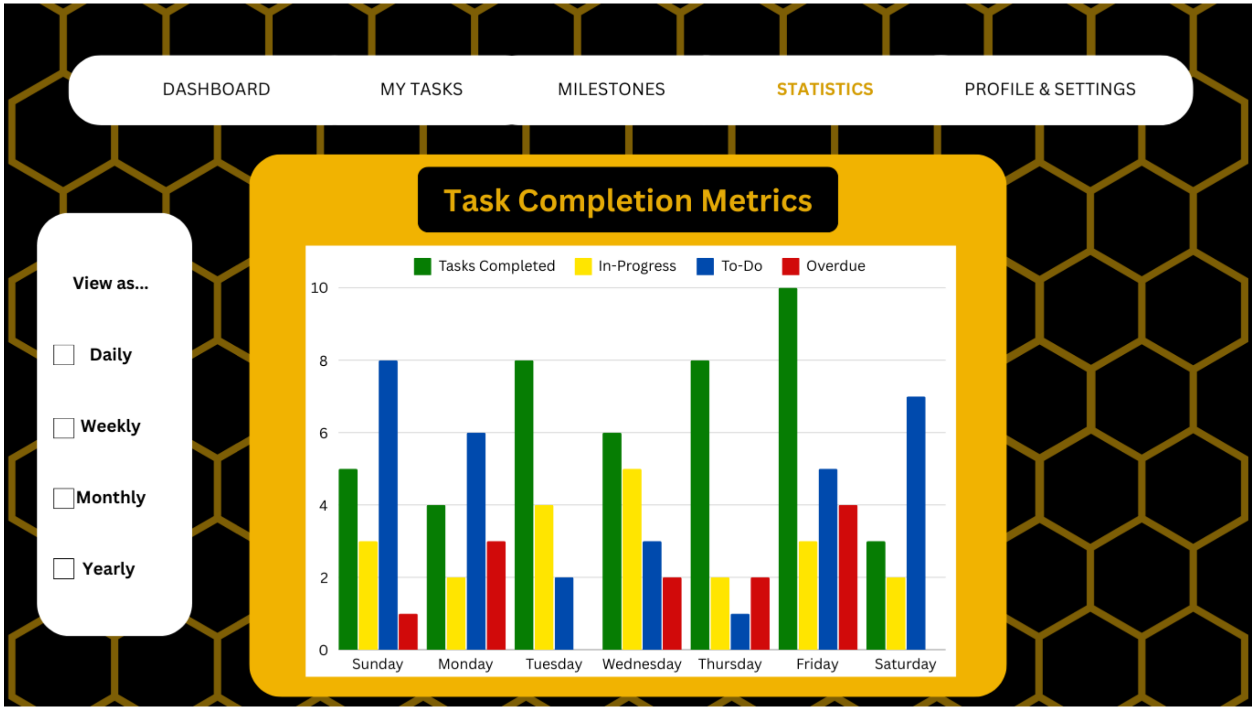

DASHBOARD
MY TASKS
MILESTONES
STATISTICS
PROFILE & SETTINGS
Task Completion Metrics
View as...
Daily
Weekly
Monthly
Yearly
Tasks Completed
In-Progress
To-Do
Overdue
10
8
6
4
2
0
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday

Figure 6

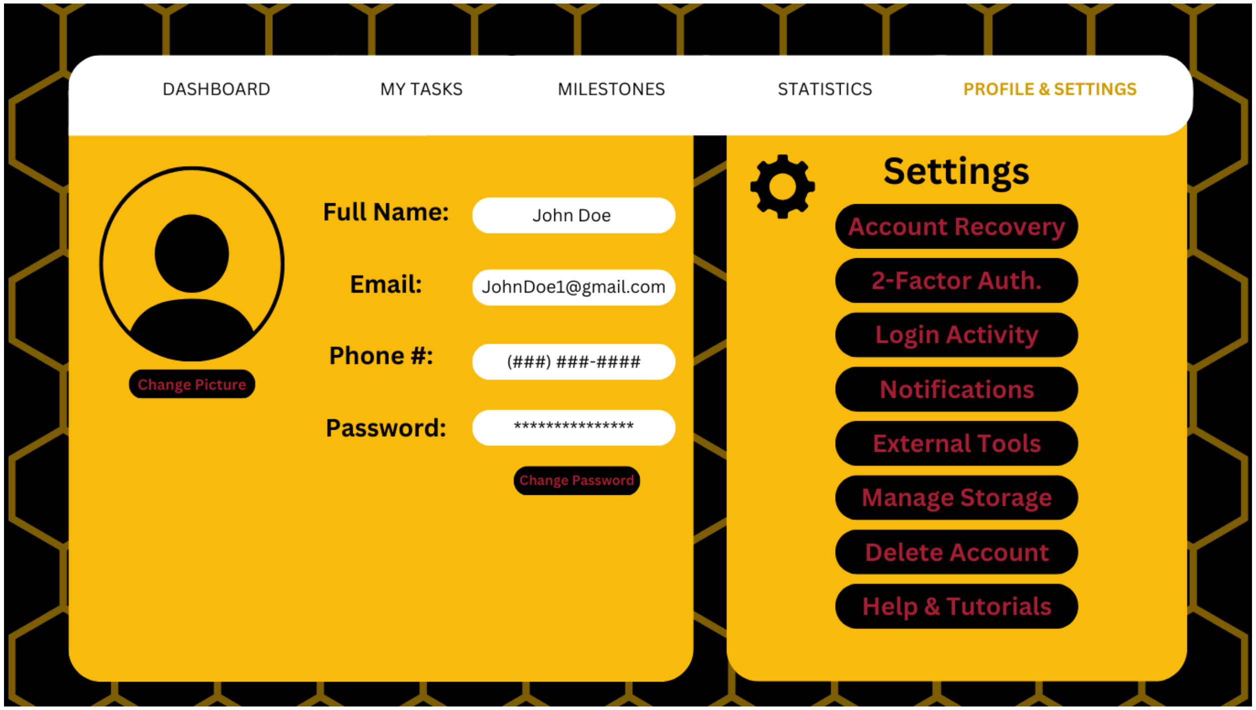

DASHBOARD
MY TASKS
MILESTONES
STATISTICS
PROFILE & SETTINGS
Full Name:
John Doe
Email:
JohnDoe1@gmail.com
Phone #:
(###) ###-####
Password:
****
Change Picture
Change Password
Settings
Account Recovery
2-Factor Auth.
Login Activity
Notifications
External Tools
Manage Storage
Delete Account
Help & Tutorials

Figure 7

Test Cases

Task Creation and Management

Unique ID	Description	Data	Expected Result	Actual Result	Pass/ Fail
TC001	Create valid Task Name	Title: “Task 1”	No error messages are shown.		
TC002	Create valid Task Description	Description: “Complete report”	No error messages are shown.		
TC003	Set valid deadline	Current Date: [11/20/2024] Due Date: [11/27/2024]	No error messages are shown.		
TC004	Select Priority Level via box selection options	Priority Level: “Select an option” ○ Low ○ Medium ○ High	No error messages are shown.		

TC005	Configure Recurrence option via checkbox	If recurrence unchecked: no action needed If recurrence checked: Select 'Recurrence Frequency' from drop list: daily, weekly, monthly	No error messages are shown.		
TC006	Store valid task to Task List via Save button	Task details (Name, description, due date, priority, recurrence selection)	Task details saved in the database with a new Task ID (e.g., TASK-0001).		
TC007	Attempt to create a task with missing name	Title: ""	Error Message: "Task Name is required."		
TC008	Create Task Name with over 20 characters	Task Name: "This is an Excessively Long Task Name"	Error: "Task name is too long"		

TC009	Create task with description with over 200 characters	Description: “[Exceeding 150 characters]”	Error Message: “Task description is too long.”		
TC010	Attempt to create task with invalid deadline	Current Date: [11/20/2024] Due Date: [11/03/2024]	Error Message: “Due Date must occur after the current date.”		
TC011	Change Task Status from the default “To-Do” Status via radio button	Task Status: “Select an option” ○ To-Do — Task created but work has not been started ○ In-Progress — Task is currently being worked on ○ Completed — Task is finished	Message: “Task Status Selected”		
TC012	Add Label to Task	Tags: "Work"	Task saved under label		

TC013	Update Task Name	Task ID: TASK-0001 New Title: “Updated Task 1”	Task updated successfully with new title		
TC014	Update task due date to future date	Task ID: TASK-001 Old Due Date: [01/10/2025] New Due Date: [01/16/2025]	Due date updated successfully		
TC015	Delete an existing task	Task ID: TASK-001	“Task deleted successfully”		
TC016	Undo deletion of a task Delete a completed task	Task ID: TASK-001	Task deletion is reversed, task restored		

Notifications and Alerts

Unique ID	Description	Data	Expected Result	Actual Result	Pass/ Fail
TC017	Enable Notifications for a user via button	User ID: 1 Notifications: On	Notifications sent to user		
TC018	Send notification for tasks due soon	Current Date: [11/20/2024] Task Due Date: [11/27/2024]	Notification: TASK DUE NEXT WEEK		
TC019	Disable notifications for a user via button	User ID: 1 Notifications: Off	No notifications sent to user		
TC020	Notification failure on server downtime	Server Down Task Created	Error Message: "Unable to send notification, server issue"		
TC021	Customize notification preference via radio button	Notification preference: “Select options” ○ Email	User will receive notifications only		

		○ SMS ○ In-App	through their selected preference.		
TC022	Set multiple reminders for a high-priority task	Task Priority: High Reminders: 1 day, 1 hour	Reminders sent at designated times		
TC023	Receive alert for high-priority task	Task Priority: High	Alert sent to user upon task creation		

User Authentication and Profile Management

Unique ID	Description	Data	Expected Result	Actual Result	Pass/ Fail
TC024	Prevent access after account deactivation	User ID: 1 (Deactivated)	User cannot log in, error message shown		
TC025	Login with valid email	Email: JohnDoe@example.com	Redirect to password input window		

TC026	Login with valid password (alphanumeric, 8+ characters, upper and lower case)	Password: aBcD123	Redirect to dashboard window		
TC027	Login with invalid email	○ Email: JohnDoe	Error Message: “Invalid email address”		
TC028	Login with blank email	Email: “”	Error Message: “Email address cannot be blank”		
TC029	Login with blank password	Password: “”	Error Message: “Password cannot be blank”		
TC030	Verify email confirmation after profile creation	Email: JohnDoe@example.com	Confirmation email sent to user with verification link		

TC031	Account lockout after multiple failed login attempts	Failed Attempts: 5	Account locked, error message shown		
TC032	Recover password using recovery option	Recovery Email: JohnDoe@example.com	Password recovery instructions sent via email		
TC033	Enable multi-factor authentication	MFA: Enabled	User prompted for secondary authentication upon login		
TC034	Encrypt all user data on storage	User Data: Task info, profile	All user data stored securely with encryption		
TC035	Auto-logout after period of inactivity	Inactivity Period: 15 minutes	User logged out automatically, prompt for re-login		

TC036	Verify secure session management for multiple logins	Login on Desktop Login on Mobile	Session managed securely without cross-device vulnerabilities		
TC037	Prevent unauthorized access to deleted account	User ID: Deleted	Error: 'Account not found or is deactivated'		
TC038	Encrypt all user data on storage	○ User Data: Task info, profile	All user data stored securely with encryption		
TC039	Auto-logout after period of inactivity	Inactivity Period: 15 minutes	User logged out automatically, prompt for re-login		
TC040	Verify secure session management for multiple logins	Login on Desktop Login on Mobile	Session managed securely without cross-device vulnerabilities		

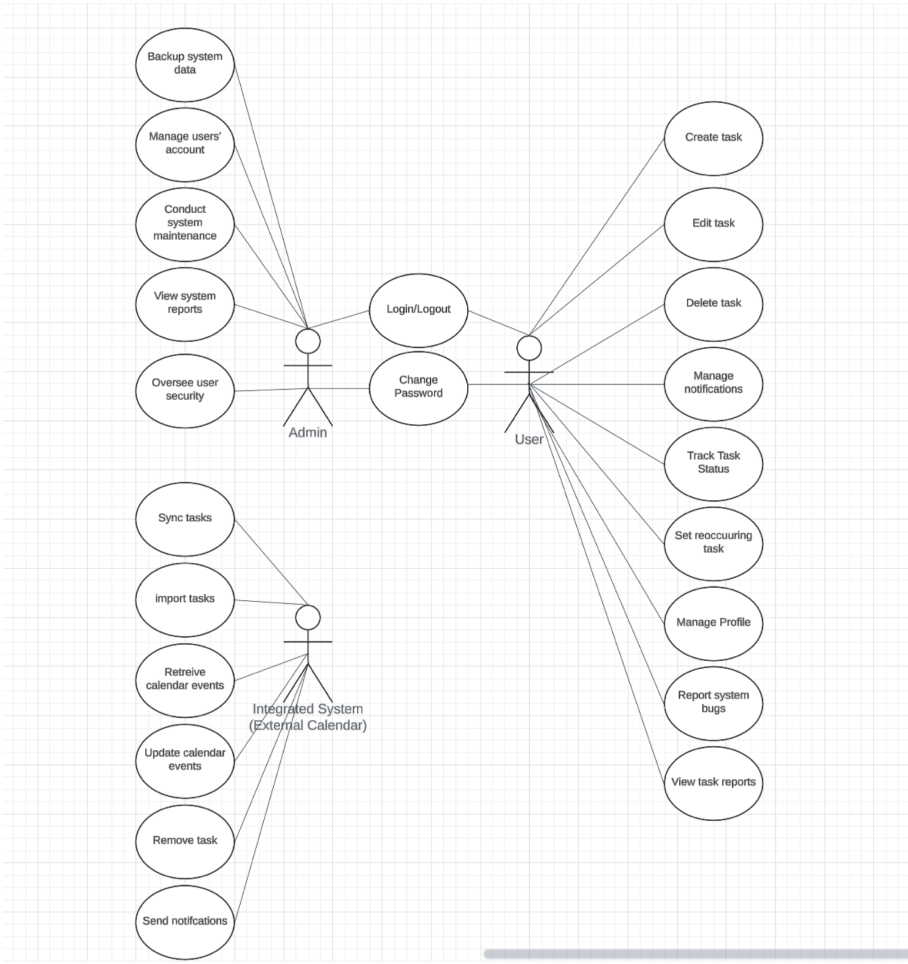
Insights and Reports

Unique ID	Description	Data	Expected Result	Actual Result	Pass/ Fail
TC041	Generate report of all tasks and their statuses	Task statuses: To-Do, In-Progress, Completed	Report generated showing tasks grouped by status		
TC042	Display productivity metrics on dashboard	Tasks: 10 Completed: 5	Dashboard shows 50% completion rate and productivity insights		
TC043	Track time spent on a specific task	Task ID: TASK-001	System logs time spent on task, time reflected in report		
TC044	Verify progress bar updates on task completion	Complete task ID: TASK-001	Progress bar on dashboard updates to reflect task completion		

Integration

Unique ID	Description	Data	Expected Result	Actual Result	Pass/ Fail
TC045	Synchronize task with external calendar and project planning applications	Calendar: Google Calendar Project planning: Microsoft Teams	Task appears in external calendar with correct details		
TC046	Sync tasks in real-time across connected devices	Device 1: Mobile Device 2: Desktop	Task changes appear instantly on both devices		
TC047	Verify integration with cloud storage for task attachments	Cloud Storage: Google Drive	Task attachments saved in Google Drive folder		

Use Case



Unique Identifier: UC1	Name: login

Actors: User, Admin
Description: The login case allows a user to login to the task management system via a valid username and password
Data: Valid Username and password
Stimulus: press the login button
Precondition: valid username and password
Postcondition: redirected to the dashboard screen
Error condition: the user enters invalid information and locked out of account
Comments

Unique Identifier: UC2	Name: Create Task
Actors: User	
Description: The Create Task case allows a user to create a task by inputting task details	
Data: Task data (name, due date, priority, description)	
Stimulus: press the create task button	

Precondition: User is authenticated and has rights to create tasks
Postcondition: redirected to the appropriate screen
Error condition: the task creation is not processed, and an error message appears
Comments

Risk Matrix

<table>
<tr><th></th><th>Highly unlikely</th><th>Unlikely</th><th>Possible</th><th>Likely</th><th>Very likely</th></tr>
<tr><td>No impact
0-100,000</td><td>Accept 1</td><td>Accept 2</td><td>Accept 3</td><td>Accept 4</td><td>Accept 5</td></tr>
<tr><td>Minor
100,000-500,000</td><td>Accept 2
R009</td><td>Accept 3
R010</td><td>ALARP 6
R011</td><td>ALARP 7</td><td>ALARP 8</td></tr>
<tr><td>Medium
500,000-1,000,000</td><td>Accept 3</td><td>ALARP 6
R001 | R005 | R012</td><td>ALARP 7
R003</td><td>ALARP 8
R007</td><td>Avoid 9</td></tr>
<tr><td>Major
1,000,000-1,500,000</td><td>ALARP 6
R002</td><td>ALARP 7</td><td>ALARP 8
R006 | R007</td><td>Avoid 9
R003 | R004</td><td>Avoid 10</td></tr>
<tr><td>Extensive
1,500,000 and up</td><td>Avoid 7</td><td>Avoid 8</td><td>Avoid 9</td><td>Avoid 10</td><td>Avoid 11
R008</td></tr>
</table>

Risk ID	Risk Category	Risk Description	Risk probability	Risk Severity	Risk Level
R001	Personnel	Poor team dynamics	40% — Unlikely	Medium	ALARP
R002	Personnel	Unclear roles and responsibilities	20% — highly unlikely	Major	ALARP
R003	Estimation	Unrealistic timeline	75% — Likely	Major	Not acceptable
R004	Estimation	Scope creep	75% — Likely	Major	Not acceptable
R005	Requirements	Complexity of requirements	40% — Unlikely	Medium	ALARP
R006	Requirements	Consistently changing requirements	60% — Possible	Major	ALARP
R007	Technology	Integration issues	60% — Possible	Major	ALARP
R008	Technology	Security vulnerabilities	80% — Very Likely	Extensive	Not acceptable
R009	Tools	Tools limitations	20%— Highly unlikely	Minor	Acceptable
R010	Tools	Dependence on third party tools	40% —Unlikely	Minor	Tolerable
R011	Usability	Accessibility issues	60% — Possible	Minor	ALARP
R012	Organizational	Resource allocation	40% — Unlikely	Medium	ALARP

Traceability

Task creation and Management

	TC 001	TC 002	TC 003	TC 004	TC 005	TC 006	TC 007	TC 008	TC 009	TC 010	TC 011	TC 012	TC 013	TC 014	TC 015	TC 016
FR 1.01	X	X	X	X	X	X	X	X	X	X		X				
FR 1.02					X											
FR 1.03												X				
FR 1.04	X	X	X	X	X								X	X		
FR 1.05															X	X
FR 1.06											X					
FR 1.07	X		X	X												
FR 1.08																
FR 1.09																

Notifcations and Alerts

	TC 017	TC 018	TC 019	TC 020	TC 021
FR 3.01	X	X		X	
FR 3.02				X	X
FR 3.03					
FR 3.04					
FR 3.05			X		

User Authentication and Profile Management

	TC 024	TC 025	TC 026	TC 027	TC 028	TC 029	TC 030	TC 031	TC 032	TC 033	TC 034
FR 4.01		X	X	X	X	X					
FR 4.02							X				
FR 4.03		X	X	X	X	X					
FR 4.04		X	X	X	X	X					X
FR 4.05											
FR 4.06									X		
FR 4.07								X			

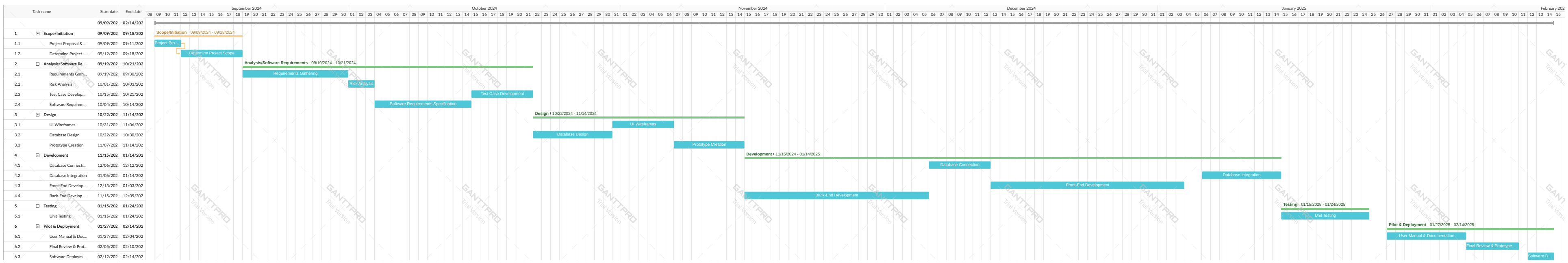
Insights and Reports

	TC 041	TC 042	TC 043	TC 044
FR 2.01	X	X		X
FR 2.02			X	

Integration

	TC 045	TC 046	TC 047
FR 5.01	X		
FR 5.02	X		X

Schedule



GANTT CHART

		Task Mode	Task Name	Duration	Start	Finish	Predecessors
1			Client Meeting	1 day	Fri 1/17/25	Fri 1/17/25	
2			Project Plan Creation	14 days	Mon 1/20/25	Thu 2/6/25	1
3			Requirements	5 days	Fri 2/7/25	Thu 2/13/25	2
4			Finalizing	2 days	Fri 2/14/25	Mon 2/17/25	3
5			Risk Assesment	2 days	Fri 2/7/25	Mon 2/10/25	2
6			Database layout	5 days	Tue 2/11/25	Mon 2/17/25	5
7			PDM Creation	7 days	Tue 2/18/25	Wed 2/26/25	6
8			UI Design	3 days	Tue 2/18/25	Thu 2/20/25	4
9			Design Phase Completion	1 day	Fri 2/21/25	Fri 2/21/25	8
10			Database Connection	4 days	Thu 2/27/25	Tue 3/4/25	7
11			Frontend Development	14 days	Wed 3/5/25	Mon 3/24/25	10
12			Backend Devlopment	28 days	Wed 3/5/25	Fri 4/11/25	10
13			Database Integration	7 days	Mon 4/14/25	Tue 4/22/25	11,12
14			Prototype Creation	3 days	Mon 4/14/25	Wed 4/16/25	11,12
15			Unit Testing	5 days	Thu 4/17/25	Wed 4/23/25	14
16			Structual Walkthrough/User Manual	5 days	Thu 4/17/25	Wed 4/23/25	14
17			Final Deliverable	6 days	Thu 4/24/25	Thu 5/1/25	15,16
18			Finishing Final Project	6 days	Fri 5/2/25	Fri 5/9/25	17